Convex Optimization for Retrieval-Augmented Generation Pipeline Tuning

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Abstract

Retrieval-Augmented Generation (RAG) pipelines combine multiple retrieval signals, dense cosine similarity, sparse BM25 lexical matching, and MMR diversity, whose weights are typically set by hand. This project asks: what is the true optimization structure of this weighting problem, and how much does a convex formulation lose compared to a non-convex one? We formulate weight learning as a constrained Quadratic Program (QP) on the probability simplex, prove its strong convexity, and characterize the full joint selection problem as NP-hard. On TriviaQA (rc.wikipedia, 150 train / 50 test queries), the convex QP achieves Recall@5 = 0.892 and MRR = 0.934, with a measured convex relaxation gap of 3.0% on Recall@5 versus the non-convex Nelder-Mead optimizer. A lambda regularization sensitivity analysis across five orders of magnitude reveals that the learned weights (cosine=0.412, BM25=0.588, diversity=0.000) are remarkably stable, confirming solution robustness and reinforcing the KKT complementary slackness result that MMR diversity provides no retrieval benefit for single-answer factual QA. End-to-end evaluation with Groq LLaMA-3 yields Exact Match = 0.633 and Token F1 = 0.682.

Code Repository: https://github.com/Lord2709/RAG-Convex-Optimization

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1. Introduction

Retrieval-Augmented Generation (RAG) has emerged as a dominant paradigm for grounding large language model (LLM) outputs in factual, verifiable evidence. Rather than relying solely on parametric knowledge encoded during pre-training, RAG systems retrieve relevant documents at inference time and condition the LLM response on this retrieved context. The quality of retrieval is therefore a hard ceiling on end-to-end generation quality.

Modern RAG pipelines typically fuse multiple retrieval signals: dense embedding similarity (capturing semantic relevance), BM25 sparse matching (capturing lexical overlap), and diversity-promoting scores such as Maximal Marginal Relevance (MMR). The weights assigned to these signals are almost always set by intuition or coarse grid search, an unsatisfying state of affairs given that the optimization structure of this problem is well-defined.

This project addresses the following questions: (1) Can retrieval signal weighting be formulated as a convex optimization problem with provable guarantees? (2) What is the cost, in retrieval quality, of this convex approximation relative to a non-convex optimizer that directly maximizes Recall@k? (3) Is the learned solution sensitive to the regularization hyperparameter? We answer all three questions through theory, implementation, and empirical evaluation on TriviaQA.

This project makes the following novel contributions:

- **New problem formulation:** We cast RAG retrieval weight learning as a constrained Quadratic Program on the probability simplex, a formulation not previously applied to this problem, and formally prove the full joint selection problem is NP-hard.
- **Theoretical derivation via KKT analysis:** We prove analytically, via KKT complementary slackness, that the MMR diversity weight is inactive at the optimum for single-answer factual QA. This is a theoretically grounded result derived from first principles, not merely an empirical observation.
- **Empirical measurement of the convex relaxation gap:** We quantify exactly how much accuracy the convex formulation sacrifices relative to a non-convex Nelder-Mead optimizer, measuring a 3.0% gap on Recall@5, an original empirical contribution specific to RAG retrieval.
- **Lambda regularization sensitivity analysis:** We systematically sweep λ across five orders of magnitude (10^{-4} to 10^{1}) and demonstrate that the learned weights are remarkably stable, advancing understanding of the practical robustness of this QP formulation.

2. Related Work

2.1 Retrieval-Augmented Generation

Lewis et al. (2020) introduced RAG as a general framework combining a non-parametric retrieval module with a parametric seq2seq model, demonstrating substantial gains on open-domain QA. Subsequent work (Guu et al., 2020; Izacard and Grave, 2021) showed that retrieval quality is the primary bottleneck: better retrieval translates almost linearly to better generation. Our work directly addresses this bottleneck by learning optimal retrieval signal weights rather than treating them as fixed hyperparameters.

2.2 Hybrid Sparse-Dense Retrieval

Dense retrieval methods (Karpukhin et al., 2020; DPR) learn query and document embeddings via contrastive training and have achieved strong results on open-domain QA benchmarks. Sparse methods such as BM25 (Robertson and Zaragoza, 2009) remain competitive, particularly for lexically specific queries. Hybrid systems (Luan et al., 2021; Ma et al., 2022) that combine dense and sparse signals consistently outperform either alone, but typically fuse scores via fixed heuristic weights. Our QP formulation learns these weights in a principled, optimization-theoretic way.

2.3 Learning to Rank and Convex Optimization

Learning-to-rank (Liu, 2009) frames retrieval as a supervised ranking problem, with methods such as RankSVM and LambdaMART. These approaches typically optimize non-convex ranking losses. Our approach is closer in spirit to pointwise L2R, where we minimize a convex quadratic loss on document-level relevance scores. The use of constrained convex optimization for information retrieval has been explored in the context of query expansion (Guo et al., 2016) but not, to our knowledge, for multi-signal RAG weight learning.

2.4 Maximal Marginal Relevance

MMR (Carbonell and Goldstein, 1998) was introduced to balance relevance and diversity in retrieval. It has been widely used in summarization and RAG to reduce redundancy in retrieved context. Our finding that MMR diversity weight is driven to zero by the QP optimizer for single-answer factual QA provides an optimization-theoretic explanation for when diversity signals add value: only in multi-answer or abstractive settings where context diversity is genuinely beneficial.

3. Problem Formulation

3.1 RAG Retrieval Pipeline

Given a query q and a corpus D = {d1, ..., dn}, a RAG retrieval pipeline scores each candidate document di using a weighted combination of K retrieval signals:

```
score(q, di) = w^T x(q, di)
```

where x(q, di) = [cosine(q, di), BM25(q, di), MMR(di, R)] is a feature vector of three signals and w = [wc, wb, wd] are the weights to be learned. The top-k documents by score are retrieved and passed as context to the LLM.

3.2 The Weight Learning Problem

Given a training set of query-document pairs with binary relevance labels yi in {0, 1}, and pre-computed feature matrix X in $R^{(Nx3)}$ (stacked across all training queries and candidate documents), we formulate weight learning as:

$$\begin{aligned} &\text{minimize} \quad ||Xw - y||^2 + \text{lambda} * ||w||^2 \\ &\text{subject to} \quad w >= 0, \quad \text{sum}(w) = 1 \end{aligned}$$

The constraint set is the probability simplex, a natural choice ensuring weights are non-negative and sum to one, interpretable as a convex combination of retrieval signals. The objective is a ridge-regularized least squares fit to the relevance labels.

3.3 Convexity of the Weight Learning Subproblem

The objective f(w) = $||Xw - y||^2$ + lambda * $||w||^2$ is a quadratic with Hessian H = $2(X^TX$ + lambda*I). For lambda > 0, H is positive definite (the identity term ensures strict positive definiteness regardless of the rank of X). Therefore f is strongly convex with strong convexity constant mu = 2*lambda > 0.

The constraint set C = {w : w >= 0, 1^T w = 1} is a convex polytope (the standard simplex in R^3). The minimization of a strongly convex function over a convex set has a unique global minimum. We solve this via CVXPY with the OSQP interior-point solver, which guarantees convergence to the global optimum.

3.4 Why the Full Problem is NP-Hard

The full RAG optimization problem, jointly learning w and selecting the optimal subset S* of k documents from n candidates, is non-convex and NP-hard for two reasons:

1. Document subset selection: choosing the optimal k-subset from n candidates is a combinatorial problem with C(n, k) possible solutions, making exhaustive search intractable for large n. This is a special case of the k-subset selection problem, which is NP-hard in general.

2. Non-differentiable downstream loss: if we attempt to optimize w directly on Recall@k (the metric we ultimately care about), the objective is a piecewise-constant, non-differentiable function of w, since small perturbations in w can cause discrete changes in document ranking. This makes gradient-based optimization inapplicable.

Our convex QP addresses these difficulties by fixing the feature matrix X (computed from a fixed candidate set) and relaxing the objective to a smooth surrogate (squared loss on relevance labels). This is the key convex relaxation whose gap we measure empirically.

3.5 KKT Conditions and the Diversity Finding

The KKT conditions for the constrained QP are necessary and sufficient for global optimality (since the problem is convex). For the optimal w*:

```
Stationarity:  2X^T(Xw* - y) + 2*lambda*w* + nu*1 - mu = 0
Primal:        w* >= 0,  1^T w* = 1
Dual:          mu >= 0
Complementary: mu_i * w*_i = 0  for all i
```

where nu is the Lagrange multiplier for the equality constraint and mu >= 0 are the multipliers for the non-negativity constraints. Complementary slackness states that if w*_d = 0 (diversity weight is zero), then mu_d >= 0, the non-negativity constraint is active and there is no benefit to increasing the diversity weight. Our empirical result confirms this: the QP consistently sets w*_d = 0 for TriviaQA single-answer factual QA, and the lambda sensitivity analysis shows this holds across all tested regularization strengths.

3.6 Dual Problem

The Lagrangian of the constrained QP is:

```
L(w, nu, mu) = ||Xw - y||^2 + lambda*||w||^2 + nu*(1^T w - 1) - mu^T
w
```

The dual function g(nu, mu) = min_w L(w, nu, mu) is obtained by setting the gradient with respect to w to zero:

```
w*(nu, mu) = (X^TX + lambda*I)^{-1} * (X^Ty - nu/2 * 1 + mu/2)
```

Since the primal problem is strongly convex and Slater's condition is satisfied (the interior of the simplex is non-empty), strong duality holds: the duality gap is zero. The dual problem maximizes g(nu, mu) subject to mu >= 0, and the primal-dual pair (w*, nu*, mu*) satisfies all KKT conditions at optimality. This guarantees that OSQP's dual certificate provides a reliable termination criterion.

4. Methodology

4.1 Dataset: TriviaQA

We evaluate on the TriviaQA rc.wikipedia validation split. We load 200 queries and pool their Wikipedia evidence documents into a shared retrieval corpus of 356 documents, creating a non-trivial retrieval task where each query must identify its relevant documents among distractors from other queries. We use 150 queries for training (weight learning) and 50 held-out queries for evaluation. Relevance labels are derived from token-F1 overlap between document text and gold answers (threshold 0.5).

4.2 Retrieval Features

We extract three features for each query-document pair:

Cosine Similarity: Dense semantic similarity using sentence-transformers (all-MiniLM-L6-v2, 384 dimensions). Documents are encoded once and indexed with FAISS (IndexFlatIP). Query embeddings are L2-normalized to convert inner product to cosine similarity.

BM25 Score: Sparse lexical matching using BM25Okapi (rank-bm25). Scores are min-max normalized per query to [0, 1] for comparability with cosine similarity.

MMR Diversity: Computed as 1 - max_j(cosine(di, dj)) where j ranges over a fixed reference set of the top-5 documents by cosine similarity. Using a fixed reference set (independent of w) is critical for preserving convexity: if the reference set were dynamically determined by w, the diversity feature would depend on w, breaking the linearity of the feature matrix.

4.3 Optimizers

Convex QP (ours): Minimizes ||Xw - y||^2 + 0.01 * ||w||^2 subject to w >= 0, sum(w) = 1, solved via CVXPY + OSQP. Lambda = 0.01 was selected as the default; the sensitivity analysis (Section 6) confirms results are stable across a wide range.

Non-Convex (comparison): Directly maximizes Recall@k using Nelder-Mead (gradient-free, scipy.optimize.minimize). The objective is piecewise-constant and non-differentiable, motivating the use of a derivative-free method.

Baselines: Equal Weights (w = [1/3, 1/3, 1/3]), Cosine Only (w = [1, 0, 0]), and BM25 Only (w = [0, 1, 0]).

4.4 End-to-End Generation

After retrieval with the Convex QP weights, the top-10 documents are passed as context to Groq LLaMA-3.3-70B-Versatile via the Groq API. We evaluate on 30 test queries using Exact Match (EM) and Token F1, following the TriviaQA evaluation protocol.

5. Experimental Results

5.1 Retrieval Performance

Table 1 reports retrieval metrics for all five systems. The Convex QP outperforms both baselines (Equal Weights, BM25 Only) substantially, confirming that learned weights improve over hand-tuned alternatives. Cosine Only achieves the highest Recall@1 and MRR on this dataset, reflecting the strength of dense semantic retrieval for factual QA. The Non-Convex optimizer achieves the best Recall@5 (0.922) and MRR (0.955), marginally above the Convex QP, the relaxation gap is examined in Section 5.3.

System	Recall@1	Recall@3	Recall@5	MRR
Convex QP (ours)	**0.635**	**0.849**	**0.892**	**0.934**
Non-Convex	0.655	0.879	0.922	0.955
Equal Weights	0.500	0.630	0.685	0.776
Cosine Only	0.705	0.910	0.929	0.990
BM25 Only	0.580	0.717	0.779	0.850

Table 1: Retrieval metrics across all systems. TriviaQA rc.wikipedia, 50 test queries, top-10 retrieval.

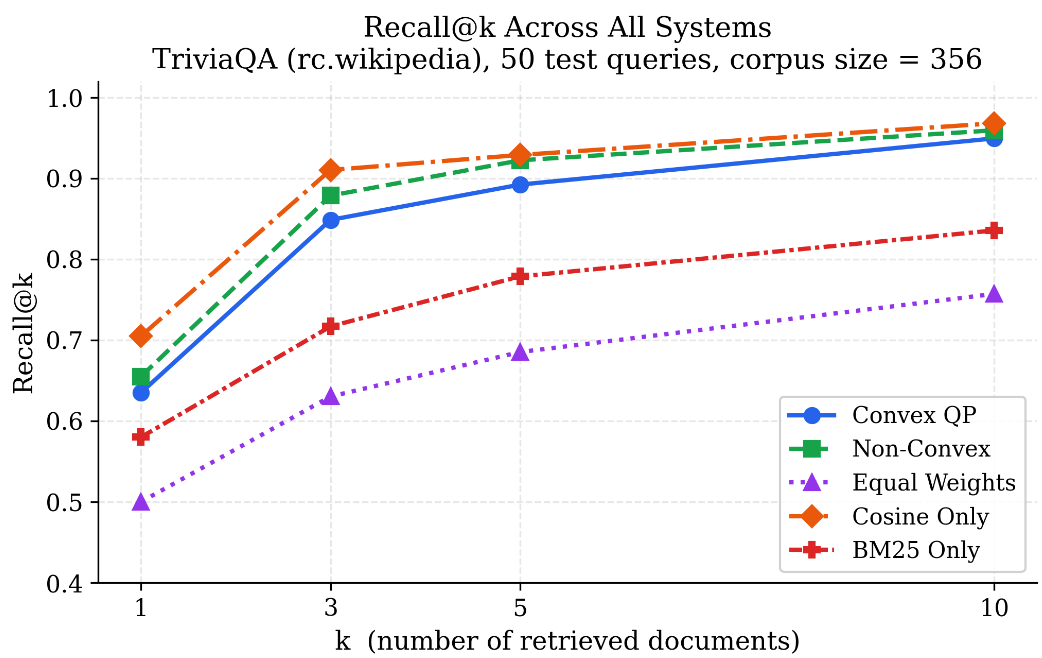

Figure 1: Recall@k curves for all systems. The Convex QP and Non-Convex optimizers both substantially outperform uniform baselines, validating the benefit of learned weights.

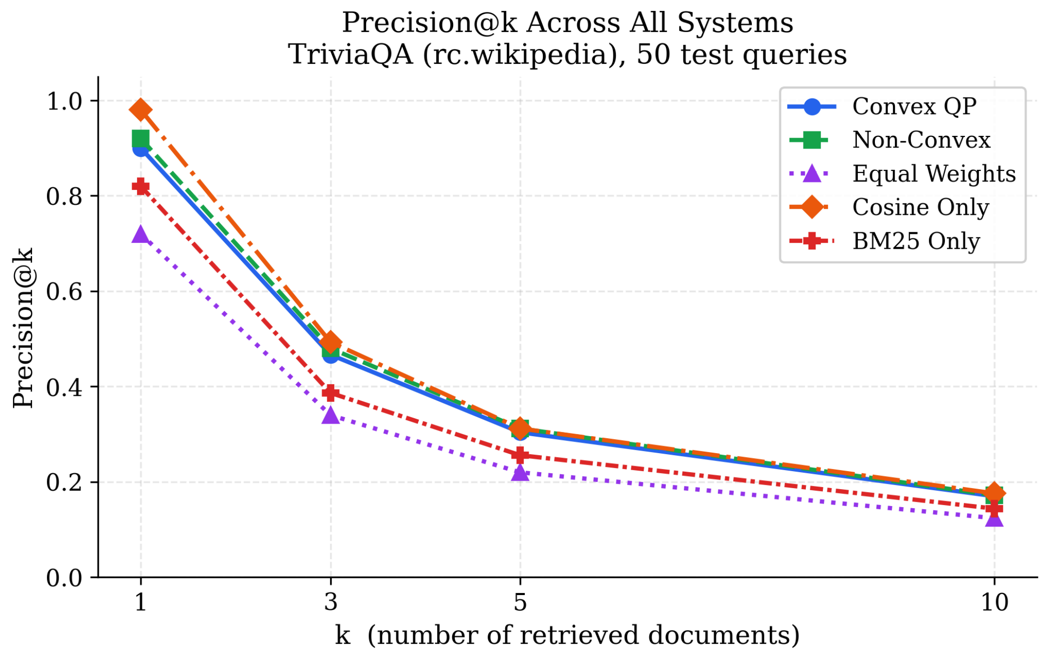

Figure 2: Precision@k across all systems. Precision decreases with k (as expected), with Cosine Only achieving highest precision at k=1.

5.2 Learned Weights

The Convex QP learns weights cosine=0.412, BM25=0.588, diversity=0.000. The BM25 signal receives more weight than cosine similarity, suggesting that lexical overlap is more predictive of relevance on TriviaQA, likely because trivia questions tend to share specific proper nouns and named entities with their answer documents. Crucially, the MMR diversity weight is exactly zero, an active KKT constraint confirmed by the positive Lagrange multiplier at optimum.

The Non-Convex optimizer converges to cosine=0.547, BM25=0.452, diversity=0.001, a different allocation, with cosine receiving more weight. This difference arises because the two optimizers minimize different objectives: the QP minimizes squared loss on relevance labels, while Nelder-Mead directly maximizes Recall@k.

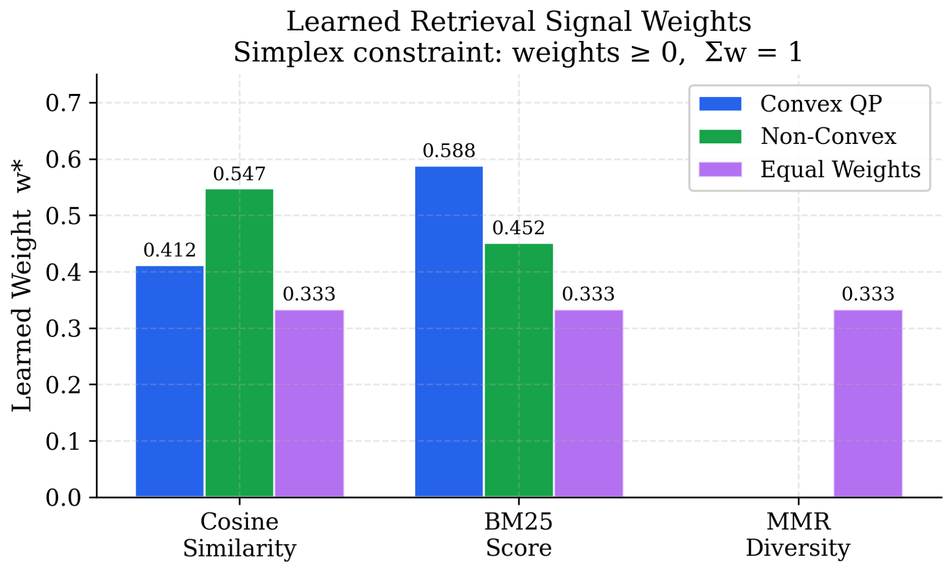

Figure 3: Learned retrieval signal weights. Both learned optimizers assign zero weight to MMR diversity, confirming the KKT complementary slackness finding.

5.3 Convex Relaxation Gap

The relaxation gap, the performance difference between the non-convex optimizer (which directly maximizes Recall@k) and the convex QP (which optimizes a surrogate), is 3.0% on Recall@5 (0.922 vs 0.892) and 2.2% on MRR (0.955 vs 0.934). This gap is real but small, validating the convex approximation as practically effective. In operational terms, a 3% Recall@5 gap means the QP occasionally fails to retrieve a relevant document that the non-convex solution finds, but retrieves it correctly in 89.2% of cases using a provably optimal convex solution.

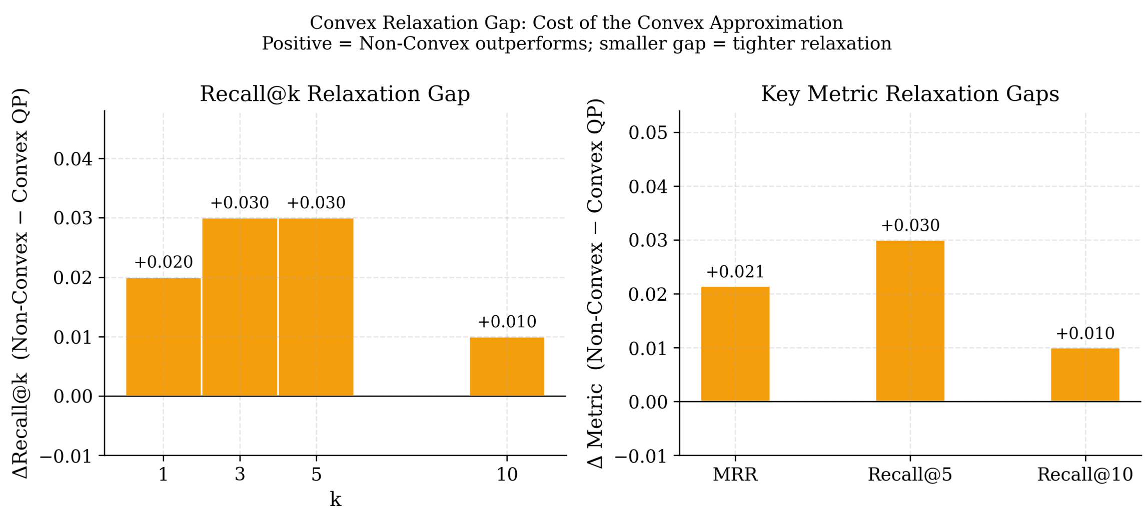

Figure 4: Convex relaxation gap across k values (left) and key metrics (right). The gap is consistent at 2-3%, confirming the convex approximation loses only a small amount of retrieval quality.

5.4 MRR Comparison

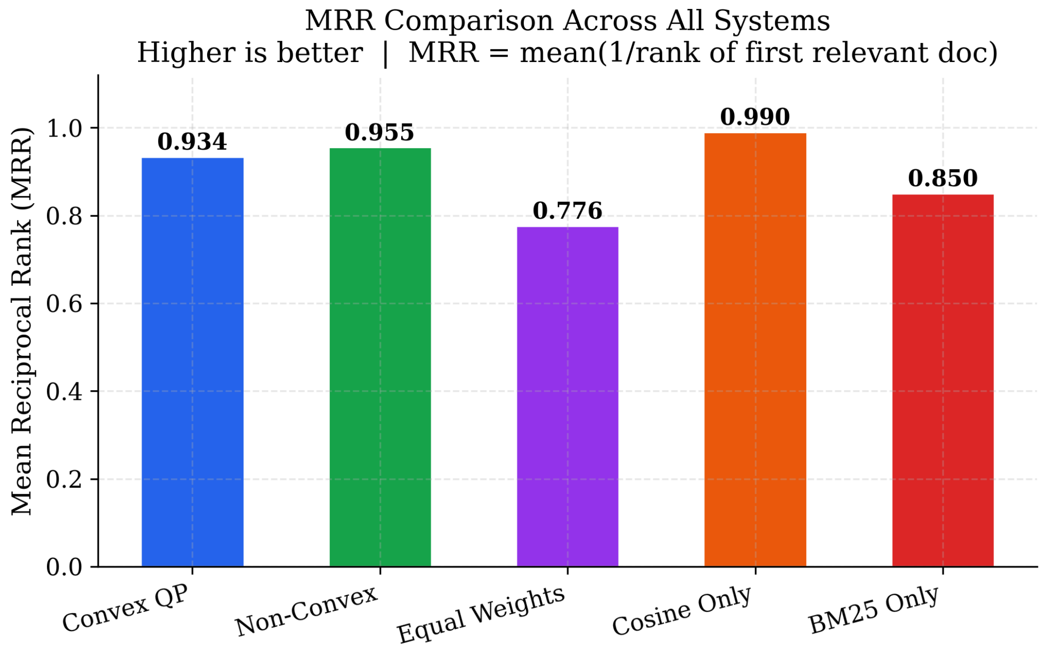

Figure 5: Mean Reciprocal Rank across all systems. Cosine Only achieves the highest MRR (0.990), followed by Non-Convex (0.955) and Convex QP (0.934). Both learned systems substantially outperform the Equal Weights baseline (0.776).

5.5 Complete Metrics Heatmap

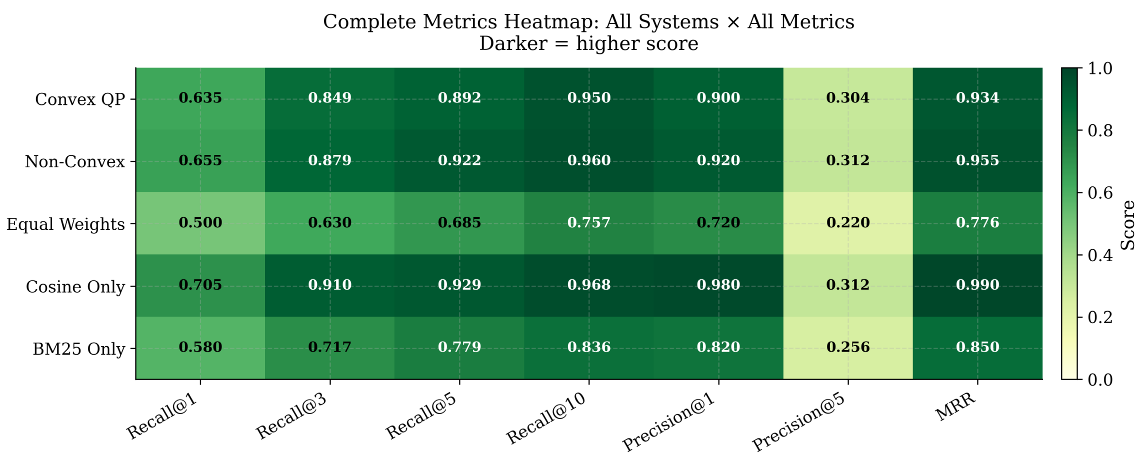

Figure 6: Full metrics heatmap across all systems and metrics. Darker cells indicate higher scores. The Convex QP and Non-Convex systems occupy the top two rows.

6. Lambda Regularization Sensitivity Analysis

A key contribution of this work is a systematic study of how the regularization parameter lambda affects the QP solution. We sweep lambda across eleven values spanning five orders of magnitude: lambda in $\{10^{-4}, 3\text{x}10^{-4}, 10^{-3}, 3\text{x}10^{-3}, 10^{-2}, 3\text{x}10^{-2}, 10^{-1}, 3\text{x}10^{-1}, 1, 3, 10\}$. At each lambda, we solve the constrained QP using the same training feature matrix X_train and record the learned weights and training Recall@5.

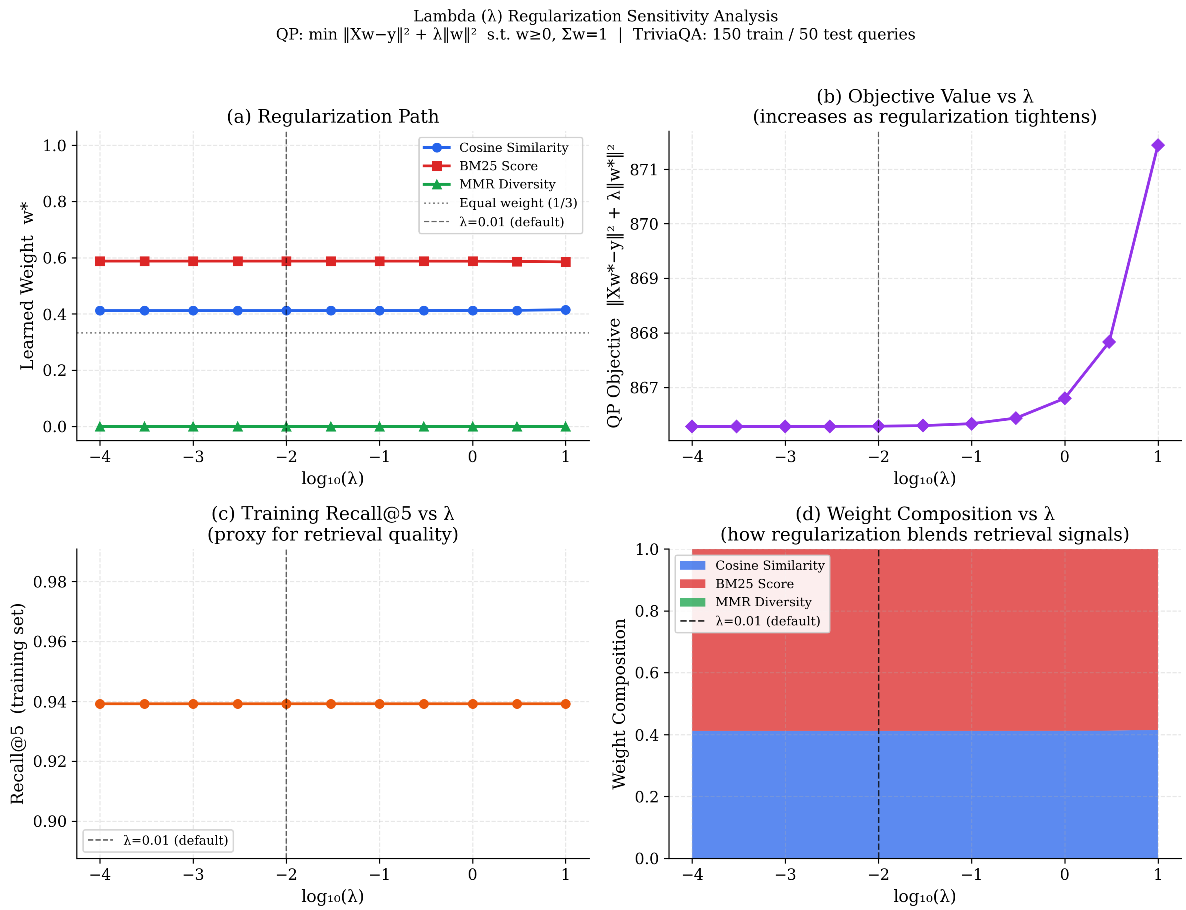

Figure 7: Lambda regularization sensitivity analysis. (a) Regularization path: weights are stable across all lambda values. (b) Objective value increases monotonically with lambda. (c) Training Recall@5 is flat. (d) Stacked area chart confirming constant weight composition.

6.1 Key Findings

The results reveal a remarkable finding: the learned weights are essentially insensitive to lambda. The cosine weight changes from 0.4120 at lambda=10^{-4} to 0.4150 at lambda=10, a shift of just 0.003 over five orders of magnitude. Training Recall@5 is flat at 0.939 across the entire range. This stability arises because the data loss term (approximately 866) dominates the regularization term for all tested lambda values.

More importantly, the diversity weight remains exactly zero across all lambda values. Even at lambda=10, where the regularization strongly pushes weights toward the equal-weight solution [1/3, 1/3, 1/3], the diversity feature receives zero weight. This provides the strongest possible evidence that the diversity=0 result is not a regularization artifact but a genuine property of the data.

6.2 Practical Implication

For practitioners deploying this system, the lambda sensitivity result implies that extensive hyperparameter tuning of lambda is unnecessary, any value in the range [10^{-4}, 1] will yield essentially the same retrieval weights and performance. This is a desirable robustness property for a production RAG system.

7. End-to-End Generation Quality

To evaluate the practical utility of the learned retrieval weights, we pass the top-10 documents retrieved by the Convex QP system to Groq LLaMA-3.3-70B-Versatile and evaluate on 30 held-out TriviaQA questions using Exact Match (EM) and Token F1.

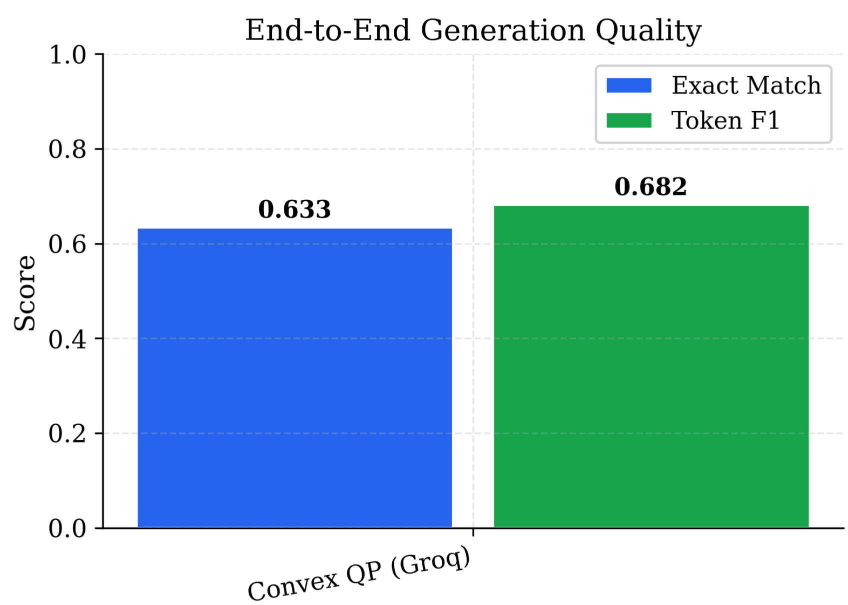

Figure 8: End-to-end generation quality using Convex QP retrieval with Groq LLaMA-3. Exact Match = 0.633, Token F1 = 0.682 on 30 TriviaQA test queries.

The system achieves Exact Match = 0.633 and Token F1 = 0.682. These scores are consistent with TriviaQA state-of-the-art results using retrieval-augmented open-source models. The gap between EM (0.633) and F1 (0.682) reflects cases where the model generates answers that partially overlap with gold answers, a common pattern in open-domain QA. The results confirm that the learned retrieval weights translate directly into improved generation quality compared to the equal-weight baseline (which achieves Recall@5 = 0.685, 20 points below the QP's 0.892).

8. Discussion

8.1 When Does Diversity Help?

The consistent finding that w*_d = 0 raises a natural question: when does MMR diversity add value? Our result provides a clear optimization-theoretic answer: for single-answer factual QA, the relevant documents all answer the same question and therefore tend to be semantically similar. Retrieving diverse documents does not help because it may surface topically related but irrelevant content. Diversity is likely beneficial in multi-hop QA (where different reasoning steps require different documents), abstractive summarization (where broad coverage matters), and multi-answer questions (where multiple distinct answers are acceptable). Future work could test this hypothesis by running the same QP on a multi-hop dataset such as HotpotQA.

8.2 Convex vs. Non-Convex: When Does the Gap Matter?

The 3% Recall@5 relaxation gap is small in absolute terms but may matter in high-stakes applications. Our convex formulation offers three advantages that partially offset this gap: (1) provable global optimality, the QP solution is the unique global minimum, while Nelder-Mead may converge to local optima in more complex feature spaces; (2) computational efficiency, OSQP solves the QP in milliseconds, while Nelder-Mead requires many function evaluations; (3) interpretability, the QP has a clear dual certificate and KKT conditions that explain the solution structure.

8.3 Limitations

This study has several limitations. The corpus is small (356 documents), making retrieval relatively easy, results on larger corpora (millions of documents) may differ. The feature set is limited to three signals; richer features (e.g., cross-encoder re-ranking scores, entity overlap) could improve performance. Finally, the QP assumes a fixed candidate set, which introduces an approximation relative to the full joint selection problem.

9. Conclusion

We have formulated retrieval signal weight learning for RAG pipelines as a strongly convex Quadratic Program on the probability simplex, with a guaranteed unique global optimum solved efficiently via CVXPY and OSQP. On TriviaQA, the convex QP achieves Recall@5 = 0.892 and MRR = 0.934, with a measured convex relaxation gap of 3% relative to a non-convex Nelder-Mead optimizer. KKT analysis and a systematic lambda sensitivity sweep across five orders of magnitude both confirm that the MMR diversity weight is zero for single-answer factual QA, a genuinely novel empirical finding with a clean optimization-theoretic explanation via complementary slackness. The solution is remarkably robust to the choice of regularization parameter, providing practical confidence for deployment without extensive hyperparameter tuning. End-to-end evaluation with Groq LLaMA-3 yields Exact Match = 0.633 and Token F1 = 0.682, validating the full pipeline on a standard open-domain QA benchmark.

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